Moral Missions: Prototyping and Designing to Support Moral Decision-Making Procedures of Responsible Data Scientists

Ask to try our prototypes !

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Introduction:

Our project contributes to a larger research study called Moral Missions, which connects how data scientists and AI practitioners navigate ethical decision-making [1] and real-world technical work [2,3].

Our portion of the project supports data scientists in translating their personal and community-based values into intentional, responsible design choices through analyzing design goals and rapid prototyping.

Research Question:

What types of interactive tools can best empower data scientists to integrate their personal moral values into concrete technical decisions?

Method:

Guided by design as research [4] practices we:

1. Generated grounded design goals from **15 interviews** with responsible data scientists
2. Rapidly prototyped and facilitated critique sessions for **30+ designs**.
3. Affinity diagrammed design goals and prototypes to derive **four key design goals for responsible data science tools**

Design Goal Examples

- *"Finding coworkers with similar values to secretly consult with"* - **P2**
- *"Documenting dataset origins, documentation gaps and assumptions"* - **P5**
- *"Communicating model uncertainty to non-technical policy makers"* - **P10**

Thematic Analysis

We thematically clustered 41 design goals to surface four themes:

1. Developing Protocols for Securing Data
2. Balancing Internal v. External Values
3. Planning, Internal Collaboration, and Resource Sharing
4. Model Evaluation, Implementation, & Data Visualization

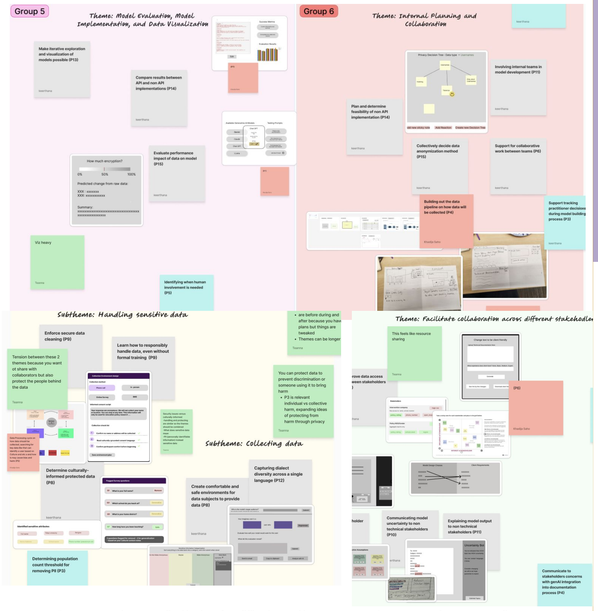
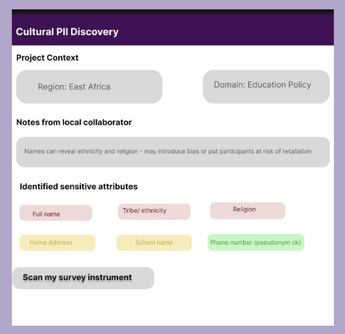


Figure 1: Screenshot of digital affinity diagramming workspace for thematically analyzing design goals and subsequent rapid prototypes.

Thematic Findings:

Developing Protocols for Securing Data



Practitioners want to create data handling protocols informed by the cultural context of PII harm

Figure 2: Prototype made for P8, a data scientist wanting to determine culturally informed protected data, to prevent data from being used for discriminatory purposes.

Balancing Internal v. External Values

Practitioners want to deliberate tradeoffs between expressing their values or supporting expectations

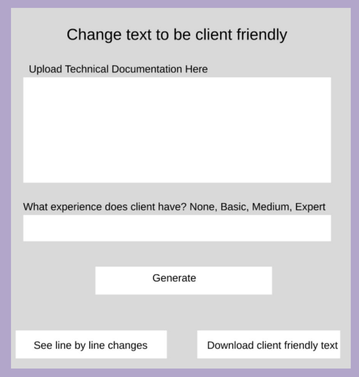


Figure 3: Prototype made for P6, a freelancer who frequently works with non-technical clients. So, the design goal was communicating technical designs to people who might not understand technical jargon.

Planning, Internal Collaboration, & Resource Sharing

Practitioners want an informed way to track and make sociotechnical decisions with colleagues

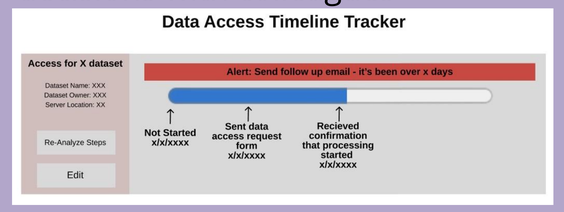
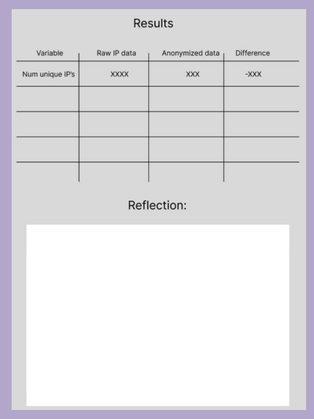


Figure 4: Prototype made for P10, an epidemiologist using EMOD, an agent based malaria modeling tool. The design goal was navigating bureaucratic barriers to accessing data.

Model Evaluation, Implementation & Data Visualization



Practitioners want to responsibly evaluate their models and visualize data with clear, reproducible, explainable, and unbiased tools

Figure 5: Prototype made for P15, a data scientist who coordinates work between different subteams. One of P15's design goals was evaluating the performance impact of encrypted data on their team's model.

Future Work:

We hope to build a responsible data science platform to:

- **Host** prototype implementations
- **Offer a space** to share and develop responsible data science techniques
- **Conduct** user testing with responsible data science practitioners

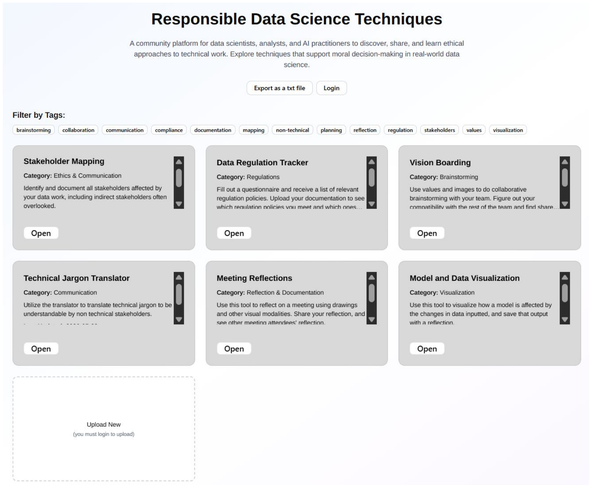


Figure 6: Screenshot of responsible data science platform mock up. We hope to implement subset of designs and test with a community of responsible practitioners.

Acknowledgements: Research was facilitated through CSE 498. Some designs utilized Figma Make, an design generation tool "powered by multiple models, including off-the-shelf foundation models and fine-tuned models"
References: Bennett (2025); [2] Kuehnert, et al (2025); [3] Deng, et al. (2022); [4] Burdick (2003)